CS2333

OBJECT ORIENTED PROGRAMMING USING JAVA

MINI PROJECT PRESENTATION

SplitEasy

Group Expense Management & Bill Splitting Application

A full-stack group expense application with an Android interface, Spring Boot REST APIs and MySQL data storage.

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MINI PROJECT PRESENTATION

The problem we are solving

Managing shared expenses in groups is often more complicated than it should be.

Scattered records

Expenses may be tracked across chats, notes, spreadsheets or memory, making the group record difficult to maintain.

Unclear balances

Members may not know who paid, who shares each cost, how much is owed or what has already been settled.

Manual calculations

Repeated calculations take time and can introduce errors, especially when expenses are shared unevenly.

SplitEasy brings group expenses and related balances into one organised application.

Project objectives

01
Group management

Create groups, manage members and organise shared activity.

02
Expense management

Add expenses, record who paid and track shared costs.

03
Splitting

Calculate each member's share of a group expense.

04
Settlements

Record repayments and reduce outstanding balances.

05
Collaboration

Support group events, polls, announcements and reminders.

06
Insights & tracking

Review spending summaries and expense history.

Key features in SplitEasy

ANDROID APPLICATION | USER INTERFACE

- User registration and login
- Create and manage groups
- Add and split expenses
- View balances and settlements
- Events, polls and announcements
- Shopping lists, reminders and chores
- Trip management and itinerary planning
- Profile, UPI details and birthday information

BACKEND SERVICES | CORE FUNCTIONALITY

- REST APIs using Spring Boot
- User, group and expense management
- Expense splitting and balance calculation
- Events, polls and trip data handling
- Reminders and notifications (planned)
- Centralised MySQL data storage
- Backend deployment and API integration

Android app ↔ REST APIs ↔ Spring Boot ↔ MySQL

How the application is structured

PRESENTATION LAYER — Android app (Java + XML) • Screens • User interaction

↓

APPLICATION LAYER — Spring Boot • REST APIs • Business logic • Expense splitting

↓

DATA ACCESS LAYER — JPA / JDBC • Repository classes • Data operations

↓

DATABASE LAYER — MySQL • Users • Groups • Expenses • Related records

Android Application ↔ REST API ↔ Spring Boot ↔ MySQL

Technology stack & core concepts

Java
Android application logic

Android (XML)
User interface layouts

Spring Boot
Backend development and REST APIs

MySQL
Relational database for persistent storage

Maven
Dependency and build management

Docker
Containerisation and deployment

OBJECT-ORIENTED PROGRAMMING

- Classes & objects — User, Group, Expense, Event and Trip entities.
- Encapsulation — keep data and operations organised within classes.
- Inheritance — use reusable base structures where applicable.
- Abstraction — separate models, services and controllers.
- Polymorphism — support common interfaces and method overrides.

Core application workflow

01 OPEN THE APP

Launch SplitEasy and access the application.

02 SELECT A GROUP

Choose the group where the expense belongs.

03 ADD AN EXPENSE

Enter description, amount, date and payer.

04 DEFINE THE SPLIT

Choose members and assign each person's share.

05 REVIEW BALANCES

Check the resulting member-wise balances.

06 RECORD SETTLEMENT

Update the record when repayments are made.

Key application modules

User & Profile

Registration, login, recovery and profile management.

Groups

Create groups and add or manage members.

Expenses

Add expenses, split amounts and track balances.

Settlements

Record payments and mark amounts settled.

Events

Create and manage group events and calendar.

Polls

Create polls and collect member votes.

Shopping Lists

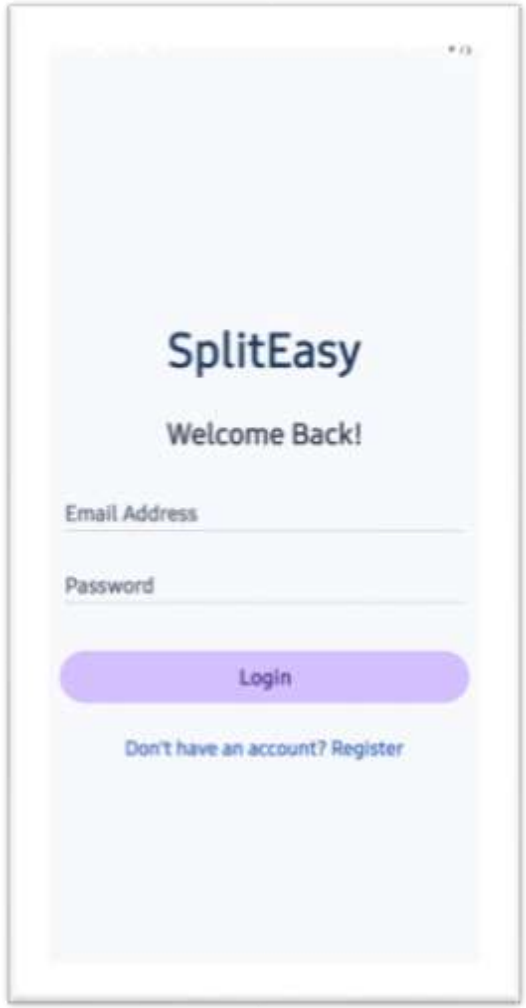
Maintain shared shopping lists and chores.

Trips

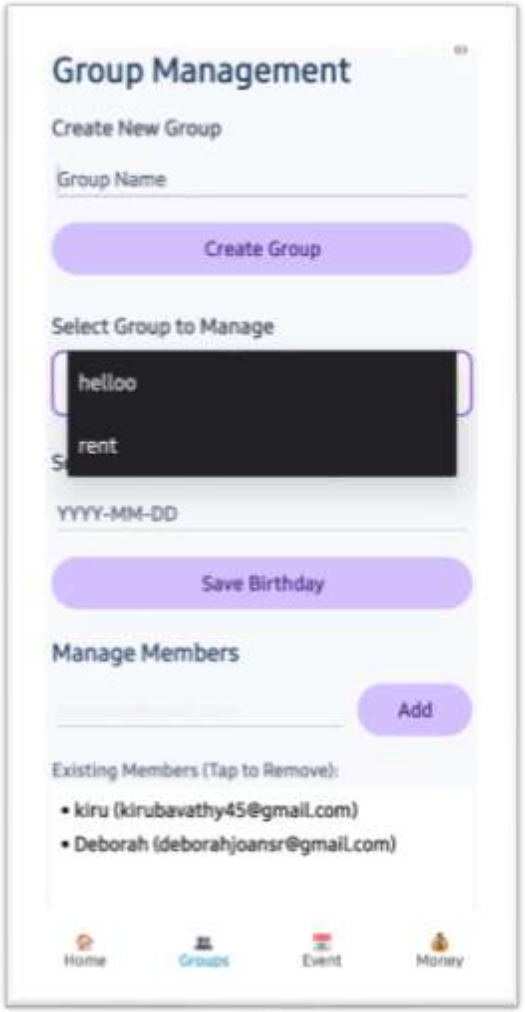
Manage trip expenses and itinerary details.

Application screens (Android)

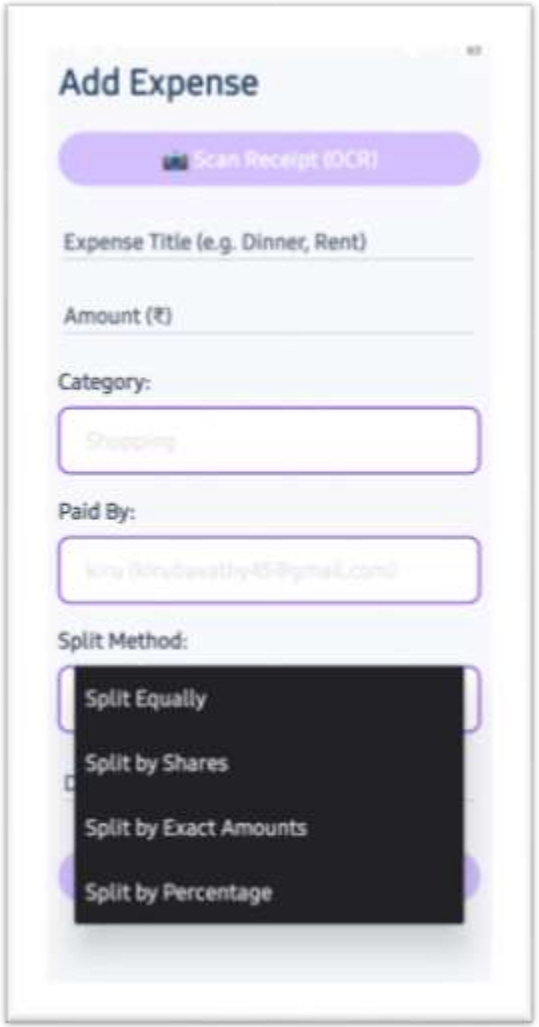
LOGIN



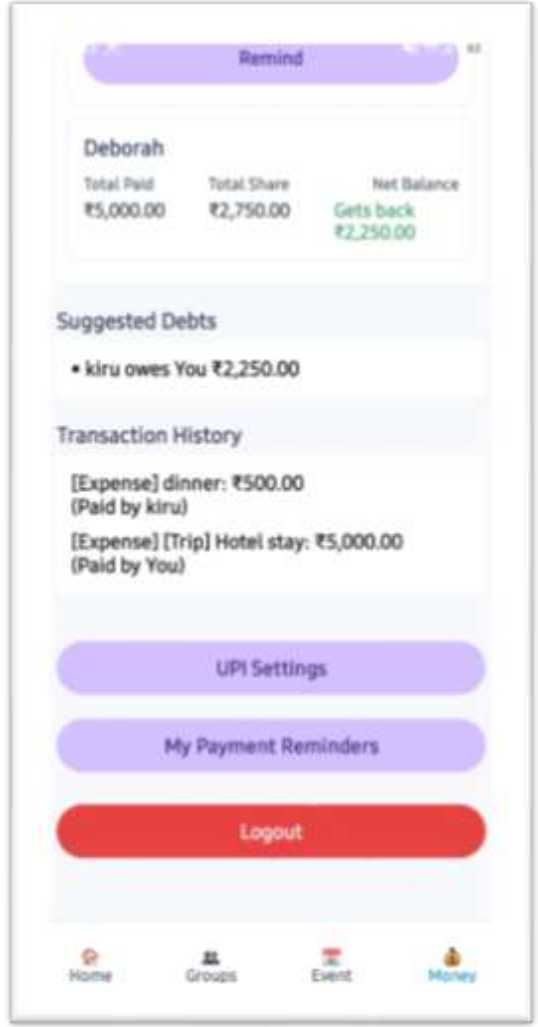
GROUPS



ADD EXPENSE



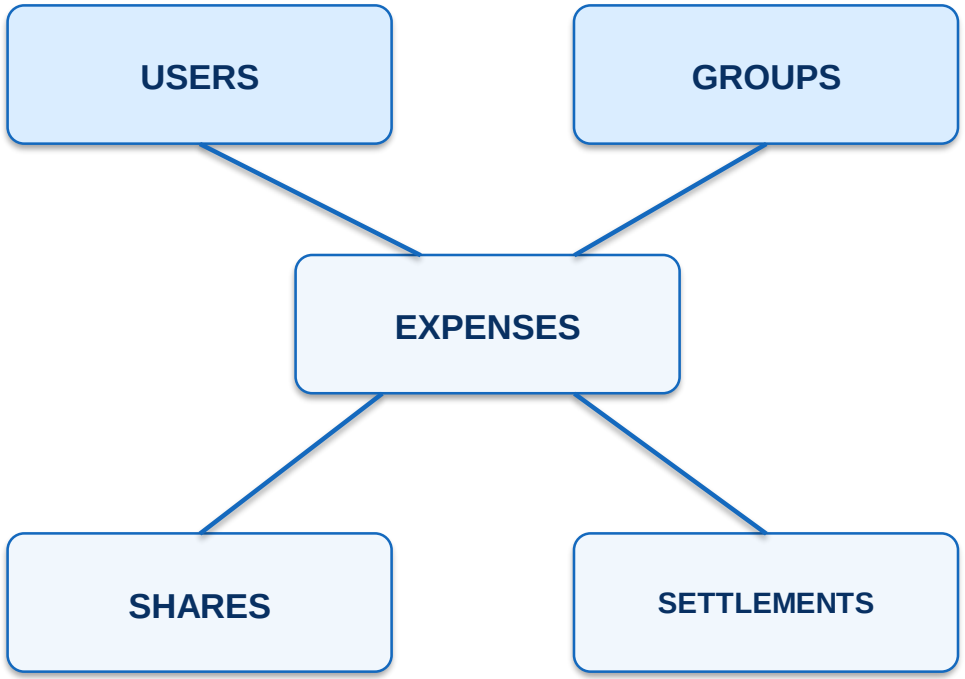
BALANCE



Database design & relationships

CORE DATA AREAS

- users — user details and account information
- groups — group information and ownership
- group_members — membership and roles
- expenses — amount, payer, date and description
- expense_shares — each member's share
- settlements — payments between members
- events / polls — group activities
- trips / reminders — itinerary and reminders



Groups organise members and expenses. Expense shares record each participant's portion; settlements track repayments.

Summary and next steps

PROJECT SUMMARY

SplitEasy is designed to help friends, roommates and travel groups manage shared expenses in one place.

The project brings together group organisation, expense entry, cost splitting, balance tracking and settlement records.

Its technical scope demonstrates Android development, Java programming, REST API integration and relational database management.

KEY TAKEAWAYS

- Addresses a common everyday problem
- Organises group spending information
- Connects an Android interface with backend services
- Provides a foundation for future features

FUTURE ENHANCEMENTS

- Complete end-to-end testing
- Stronger authentication and authorisation
- Notifications for pending settlements
- Exportable expense reports
- Offline access and synchronisation
- Improved validation and reliability

THANK YOU!

SPLITEASY